

Software Requirements Specification (SRS)

Project: Smart Warranty & Service Vault System

1. Introduction

A web application to securely manage product warranties, invoices, manuals, warranty cards, service history and reminders.

2. Objectives

Centralized warranty storage, expiry tracking, service history, OCR invoice extraction, notifications, secure authentication.

3. Users

User (Customer), Administrator.

4. Functional Requirements

Registration/Login, JWT auth, CRUD Products, Upload Documents, Warranty Tracking, Service History, OCR auto-fill, Email reminders, Dashboard, Reports.

5. Non-Functional Requirements

Security, scalability, maintainability, responsive UI, performance, availability.

6. Technology Stack

Frontend: React.js; Backend: Spring Boot(Java21); Database: PostgreSQL; ORM: Spring Data JPA/Hibernate; Security: Spring Security+JWT; OCR:Tesseract; Notifications: Spring Scheduler + Email.

7. Database Tables

users, products, warranties, documents, service_records, notifications.

8. External APIs

Tesseract OCR, SMTP Email.

9. Future Enhancements

Barcode scanning, cloud backup, mobile app, AI warranty assistant.